Bar-to-bar matchings for monomorphisms and epimorphisms

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Abstract

Using an algebraic decomposition of the space of morphisms between pointwise finite-dimensional persistence modules, we induce, from monomorphisms and epimorphisms of persistence modules, matchings between the modules' barcodes. Associating to these matchings so-called p-costs, we are able to algebraically define p-Wasserstein distances and prove their equivalence to the combinatorially defined distance. Extending our induced matchings to arbitrary morphisms by the epi-mono factorization strategy of Bauer and Lesnick (2015), we also recover the partial additive matching of Gonzalez-Diaz, Sorriano-Trigueros, and Torras-Casas (2025).

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Contents

1 Introduction

In this paper, we generalize the results on algebraic Wasserstein distances proven for tame persistence modules in [1] to pointwise finite-dimensional persistence modules. In this way, we recover the results of [14].

The approach of this paper is more robust than that of [1], casting bar-to-bar morphism as the main character.

After covering preliminaries in Section 2, we introduce bar-to-bar morphisms and their induced matchings in Section 3. We conclude the paper with Section 4, where we recover the isometry result for Wasserstein distances.

2 Preliminaries

2.1 Persistence modules

We fix a field K and denote by vect_K the category of finite dimensional vector spaces over K. Viewing $(\mathbb{R}, \leq)$ as a poset category, a **pointwise finite dimensional persistence module over** K is a functor $X\colon \mathbb{R} \to \mathrm{vect}_K$, consisting therefore of a collection of finite dimensional vector spaces $\{X_t\}_{t\in\mathbb{R}}$ and linear maps between them $X_{s\leq t}$ for $s \leq t$, we call transition functions. We denote the category of pointwise finite dimensional persistence modules and natural transformations between them by Pfd. In the following, pointwise finite dimensional persistence modules are referred to as persistence modules for brevity. The category Pfd has biproducts (i.e. direct sums), which are computed pointwise.

In this setting, interval modules are not parameterized by the real line, but rather by an extension of the real line:

▶ **Definition 2.1** (Decorated real line and intervals). *This definition is adapted from [3, §2.3]. We consider the* ***decorated real line***

$$\mathbb{E} \coloneqq \mathbb{R} \times \{-,+\} \cup \{(-\infty,+),(+\infty,-)\} \subseteq [-\infty,+\infty] \times \{-,+\}$$

equipped with the following lexicographic order: for (s,σ) and (t,τ) in $\mathbb{E}$, we have $(s,\sigma) < (t,\tau)$ if $s < t$ in $[-\infty,+\infty]$ or if $s = t$, $\sigma = -$, and $\tau = +$. We denote $(t,-)$ and $(t,+)$ in $\mathbb{E}$ by t^- and t^+, respectively, except when $t = \pm\infty$, where we omit the sign.

Ordered pairs of elements of $\mathbb{E}$ are in one-to-one correspondence with all intervals on the real line: for $s^\sigma < t^\tau$ in $\mathbb{E}$, we define the interval

$$\langle s^\sigma, t^\tau\rangle \coloneqq \{u \in \mathbb{R} \mid s < u < t \text{ or } s^\sigma = u^- \text{ or } t^\tau = u^+\}.$$

In particular, for $s < t$ in $\mathbb{R}$, we have the following equality of intervals:

$$\langle s^-, t^-\rangle = [s,t), \qquad \langle s^-, t^+\rangle = [s,t],$$
$$\langle s^+, t^-\rangle = (s,t), \qquad \langle s^+, t^+\rangle = (s,t].$$

▶ **Definition 2.2** (Overlapping intervals). *Let $a,b,c,d \in \mathbb{E}$. Following [4], we say that* ***the interval*** $\langle a,b\rangle$ ***overlaps the interval*** $\langle c,d\rangle$ ***above*** *(and antisymmetrically, $\langle c,d\rangle$* ***overlaps*** *$\langle a,b\rangle$* ***below****) if $c \leq a < d \leq b$.*

▶ **Definition 2.3** (Interval modules). *Let $a < b$ be elements of $\mathbb{E}$. A* ***bar*** *(or* ***interval module****) with* ***start-point*** *a and* ***end-point*** *b, denoted as $K(a,b)$, is the persistence module defined as follows: for any u in $\mathbb{R}$,*

$$K(a,b)_u \coloneqq \begin{cases} K & \text{if } u \in \langle a,b\rangle,\\ 0 & \text{otherwise,}\end{cases}$$

and for any $u \leq v$ in $\mathbb{R}$,

$$K(a,b)_{u\leq v} \coloneqq \begin{cases} \mathrm{id}_K & \text{if } u,v \in \langle a,b\rangle,\\ 0 & \text{otherwise.}\end{cases}$$

In particular, the ***support*** *of $K(a,b)$, i.e. the real values where the functor is nonzero, is the interval $\langle a,b\rangle$.*

▶ Remark 2.4. We observe that there exists a nonzero morphism $K(a,b) \to K(c,d)$ if, and only if, $\langle a,b\rangle$ overlaps $\langle c,d\rangle$ above.

In order to better manage these interval modules, we will be considering the following subcategory of Pfd. Recall that a **multiset** is a set S equipped with a **multiplicity** $m_S \colon S \to \mathbb{N}$, and that when there is no ambiguity, S and m_S may be used interchangeably.

▶ **Definition 2.5** (Subcategory of bars)**.** *We denote by* Bar *the full subcategory of* Pfd *whose objects are the (potentially infinite) direct sums of bars $\bigoplus_{i\in I} K(a_i,b_i)$. We call this the* ***category of bars****.*

Given an object $X = \bigoplus_{i\in I} K(a_i,b_i)$ of Bar*, we define its* ***barcode*** *as the multiset $\mathcal{B}(X) \coloneqq \{\langle a_i,b_i\rangle\}_{i\in I}$ of intervals appearing in X.*

▶ **Definition 2.6.** *Let X be a persistence module in* Pfd*. A* ***barcode decomposition*** *of X is the given of an object $\bigoplus_i K(a_i,b_i)$ in* Bar *and an isomorphism $\varphi_X \colon X \cong \bigoplus_i K(a_i,b_i)$.*

▶ Remark 2.7. The decomposition theorem [9, 5] states that every persistence module X in Pfd admits a barcode decomposition $\varphi_X \colon X \cong \bigoplus_i K(a_i,b_i)$. While the isomorphism φ_X is not unique, the direct sum of bars $\bigoplus_i K(a_i,b_i)$ is, up to reordering of the bars. In particular, we can speak of the **barcode** of an object in Pfd as well.

▶ Remark 2.8. Every persistence module in Pfd has at most countably many bars $K(s,t)$ with $s < t \in E$ in its barcode decomposition. In other words, up to an **ephemeral** summand (see [6]), which is a direct sum of bars of the form $K(s^-,s^+)$ supported on $\{s\}$, every persistence module X in Pfd has at most countably many bars. Moreover, for each $s \in \mathbb{R}$, X has a finite number of bars $K(s^-,s^+)$, since the dimension of X_s is finite. As a consequence, for all $b \in \mathbb{E}$, X only has countably many bars with start-point b, and likewise countably many with end-point b.

▶ **Proposition 2.9.** *The categories* Bar *and* Pfd *are equivalent, given the axiom of choice.*

Proof. We observe that the inclusion functor Bar $\hookrightarrow$ Pfd is fully faithful as the inclusion of a full subcategory and is essentially surjective by the decomposition theorem. Using the axiom of choice, we conclude that the inclusion functor is an equivalence of categories: see [15, Lemma 02C3]. ◀

Recall from [1] that a persistence module X in Pfd is **tame** if there exist real numbers $0 = t_0 < t_1 < \cdots < t_k$ such that the transition function $X_{s\leq t}$ is a non-isomorphism only if $s < t_i \leq t$ for some $i \in \{1,\dots,k\}$. All the bars in the barcode decompositon of a tame persistence module are of the form $\langle s^-,t^-\rangle = [s,t)$.

As a notion of *size* for persistence modules, we use the p-norm first introduced in [7] and [14], also studied in [1] for tame persistence modules.

▶ **Definition 2.10.** *For $p \in [1,\infty]$, the p-**norm** of a persistence module with barcode decomposition $X \cong \bigoplus_i K(a_i,b_i)$ is:*

$$\|X\|_p = \begin{cases} \left(\sum_i (b_i - a_i)^p\right)^{\frac{1}{p}} & \text{for } p \in [1,\infty), \\ \sup_i\{(b_i - a_i)\} & \text{for } p = \infty. \end{cases}$$

By Remark 2.8, the p-norm of a persistence module in Pfd involves an infinite sum with at most countably many positive summands. Following [14] (Definition 2.11) we say that a persistence module X has *bounded p-norm* if $\|X\|_p^p < \infty$.

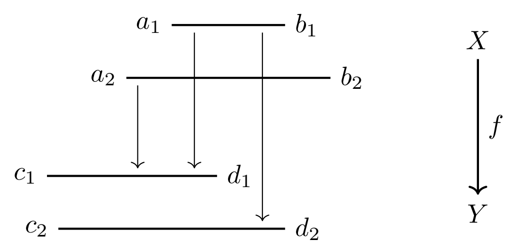


Figure 1 Representation of a morphism $f\colon K(a_1,b_1)\oplus K(a_2,b_2)\to K(c_1,d_1)\oplus K(c_2,d_2)$ in Bar. An arrow between two bars $K(a_i,b_i)$ and $K(c_j,d_j)$ corresponds to the projection of f to $\hom(K(a_i,b_i),K(c_j,d_j))$; the morphism is represented by its nonzero projections.

2.2 Morphisms and matrices

One advantage of Bar over Pfd is that we can conveniently, and not just up to isomorphism, represent morphisms as (possibly infinite) matrices. Let $X=\bigoplus_{i\in I}K(a_i,b_i)$ and $Y=\bigoplus_{j\in J}K(c_j,d_j)$ be objects of Bar. Note that $\hom(X,Y)$ is a vector space, possibly infinite-dimensional. Moreover, given the canonical isomorphisms

$$\hom\Big(\bigoplus_{i\in I}K(a_i,b_i),\bigoplus_{j\in J}K(c_j,d_j)\Big)\cong\prod_{i\in I}\prod_{j\in J}\hom(K(a_i,b_i),K(c_j,d_j)) \tag{1}$$

and

$$\hom(K(a_i,b_i),K(c_j,d_j))\cong\begin{cases}K & \text{if } \langle a_i,b_i\rangle \text{ overlaps } \langle c_j,d_j\rangle \text{ above,}\\ 0 & \text{otherwise,}\end{cases} \tag{2}$$

we can then represent each morphism $f\in\hom(X,Y)$ with a matrix $F\in K^{I\times J}$, where $F_{ij}=0$ if $\langle a_i,b_i\rangle$ does not overlap $\langle c_j,d_j\rangle$ above.

▶ **Definition 2.11** (Matrix representations of bar morphisms). *Let $f\colon\bigoplus_{i\in I}K(a_i,b_i)\to\bigoplus_{j\in J}K(c_j,d_j)$ be a morphism in* Bar*. An **ordered matrix representation of** f is $(F_{ij})_{i\in I,j\in J}$ where, for all $i\in I$ and $j\in J$, the entry F_{ij} is the projection of f to $\hom(K(a_i,b_i),K(c_j,d_j))$ via the canonical isomorphisms* (1) *and* (2)*, and I and J are ordered in a way compatible with the product order on the pairs (a_i,b_i) and (c_j,d_j).*

▶ **Example 2.12.** For example, the only ordered matrix representation of the morphism represented in Figure 1 is:

$$F=\begin{array}{c c} & \begin{array}{cc} K(a_1,b_1) & K(a_2,b_2)\end{array}\\ \begin{array}{c}K(c_1,d_1)\\ K(c_2,d_2)\end{array} & \begin{pmatrix} 1 & 1\\ 1 & 0\end{pmatrix}\end{array}$$

Under certain conditions, the composition of morphisms can also be represented by matrices.

▶ **Definition 2.13** (Column-finite matrices). *A matrix $M\in K^{I\times J}$ with rows indexed by I and columns indexed by J is **column-finite** if each column only has a finite number of nonzero coefficients.*

▶ **Proposition 2.14.** *Let $f\colon X\to Y$ and $g\colon Y\to Z$ be morphisms in* Bar*, F and G their matrix representations. Suppose that F and G are column-finite. Then the matrix* *representation of $g \circ f$ is*

$$\left(\sum_{j \in J} F_{ij} G_{j\ell} \mathbb{I}\{a_i \leq e_\ell < b_i \leq f_\ell\} \right)_{i \in I, \ell \in L},$$

where $X = \bigoplus_{i \in I} K(a_i, b_i)$, $Y = \bigoplus_{j \in J} K(c_j, d_j)$, $Z = \bigoplus_{\ell \in L} K(e_\ell, f_\ell)$, and $\mathbb{I}\{a_i \leq e_\ell < b_i \leq f_\ell\} = 1$ if the inequalities hold and 0 otherwise.

Proof. The proof is left to the reader, for now. ◀

▶ Remark 2.15.
1. The reason we assume that F and G are column-finite is to ensure that the sums are well-defined. There are other settings where the sums are well-defined, e.g. for row-finite matrices, but we will only need the result for column-finite matrices.
2. When every interval overlaps every other interval, composition of functions corresponds exactly to matrix multiplication. For example, this is the case when every interval has the same endpoint.

2.3 Matchings

Set-theoretic **matchings** are defined as partial injective functions between sets [4], but we will use the following equivalent definition in order to more easily generalize to multiset matchings.

▶ **Definition 2.16** (Matchings). *Let I and J be sets. A **(set-theoretic partial) matching between I and J** is a subset $\mu \subseteq I \times J$ such that the projections π_I and π_J onto I and J, respectively, are injective. We denote by $\mathsf{Match}(I, J)$ the set of matchings between I and J.*

We use the definitions of the category **Mch** (including composition of matchings $\circ$), **multiset representations**, **barcodes**, and the category **Barc** (including the overlap composition of matchings $\bullet$) from [4]. We also define multiset matchings:

▶ **Definition 2.17** (Multiset matchings [11, Definition 2.4]). *Let (S, m_S) and (T, m_T) be two multisets. A **(partial multiset) matching between S and T** is a multiset (M, m_M) where $M = S \times T$ and, for all $s \in S$ and $t \in T$,*

$$\sum_{t' \in T} m_M(s, t') \leq m_S(s) \quad \text{and} \quad \sum_{s' \in S} m_M(s', t) \leq m_T(t).$$

We denote by $\mathsf{Match}(S, T)$ the set of matchings between S and T.

In particular, this definition coincides with the set-theoretic one when S and T are sets (m_S and m_T bounded above by 1).

▶ **Definition 2.18** (Matching costs). *Let S and T be multisets of intervals of the real line, $M \in \mathsf{Match}(S, T)$ a matching, and $p \in [1, \infty]$. The **p-cost of M** is*

$$c_p(M) \coloneqq \left(\sum_{(\langle s^\sigma, t^\tau \rangle, \langle s'^{\sigma'}, t'^{\tau'} \rangle) \in M} m_M(\langle s^\sigma, t^\tau \rangle, \langle s'^{\sigma'}, t'^{\tau'} \rangle)(|s - s'|^p + |t - t'|^p) \right)^{\frac{1}{p}}.$$

In other words, $c_p(M)$ is the p-norm of the vector of lengths of intervals in the symmetric differences of matched intervals in M.

2.4 Dualization of persistence modules and matchings

In this paper, we mainly prove results for monomorphisms in Pfd. Analogous results hold for epimorphisms by a duality argument. Specifically, we consider the contravariant dualization endofunctor $(\cdot)^*\colon \mathsf{Pfd} \to \mathsf{Pfd}$ defined by [3, Section 3.4], which sends a map $f\colon X \to Y$ to the **dual map** $f^*\colon Y^* \to X^*$. In particular, this functor sends epimorphisms to monomorphisms, and vice versa.

The dual of a matching switches the domain and codomain, switches start- and end-points, and changes the sign of start- and end-points.

3 Bar-to-bar morphisms and induced matchings

3.1 Bar-to-bar morphisms

▶ **Definition 3.1.** *A morphism $f\colon X \to Y$ of persistence modules in* Bar *is* ***strictly bar-to-bar*** *if, writing $X = \bigoplus_{i\in I} K(a_i, b_i)$ and $Y = \bigoplus_{j\in J} K(c_j, d_j)$, there exists a (set-theoretic) matching $\mu \in \mathsf{Match}(I, J)$ such that*

$$f = \bigoplus_{(i,j)\in\mu} f_{i,j} \oplus \bigoplus_{i\in I\setminus\pi_I(\mu)} g_i \oplus \bigoplus_{j\in J\setminus\pi_J(\mu)} h_j, \tag{3}$$

where each f_{ij} is a nonzero morphism $K(a_i, b_i) \to K(c_j, d_j)$, each g_i denotes the zero morphism $K(a_i, b_i) \to 0$ and each h_j denotes the zero morphism $0 \to K(c_j, d_j)$. The matrix associated to a strictly bar to bar morphism has at most one non zero entry for each column and one non zero entry for each row.

A morphism $f\colon X \to Y$ in Pfd *is* ***bar-to-bar*** *if there exist barcode decompositions of X and Y such that the associated morphism in* Bar *is strictly bar-to-bar. In other words, there exist barcode decompositions $(\varphi_X, \bigoplus_i K(a_i, b_i))$, $(\varphi_Y, \bigoplus_j K(c_j, d_j))$ of X and Y respectively and $g\colon \bigoplus_i K(a_i, b_i) \to \bigoplus_j K(c_j, d_j)$ strictly bar-to-bar such that the following diagram commutes:*

$$\begin{tikzcd}
\bigoplus_i K(a_i, b_i) \arrow[r, "\varphi_X"] \arrow[d, "g"'] & X \arrow[d, "f"] \\
\bigoplus_j K(c_j, d_j) \arrow[r, "\varphi_Y"'] & Y
\end{tikzcd}$$

Not every morphism is bar-to-bar: see Figure 2 for an example where no isomorphisms would make $f\colon X \to Y$ strictly bar-to-bar.

▶ **Proposition 3.2.** *The composition of two strictly bar-to-bar monomorphisms is strictly bar-to-bar.*

Proof. Consider strictly bar-to-bar morphisms $f\colon X \to Y$ and $f'\colon Y \to Z$ with decompositions

$$f = \bigoplus_{(i,j)\in\mu} f_{ij} \oplus \bigoplus_{i\in I\setminus\pi_I(\mu)} g_i \oplus \bigoplus_{j\in J\setminus\pi_J(\mu)} h_j \quad \text{and} \quad f' = \bigoplus_{(j,\ell)\in\mu'} f'_{j\ell} \oplus \bigoplus_{j\in J\setminus\pi_J(\mu')} g'_j \oplus \bigoplus_{\ell\in L\setminus\pi_L(\mu)} h'_\ell,$$

where in particular $\mu \in \mathsf{Match}(I, J)$ and $\mu' \in \mathsf{Match}(J, L)$ are set-theoretic matchings. Their composition $f' \circ f$ has the decomposition

$$f' \circ f = \bigoplus_{\substack{(i,\ell)\in\mu'\circ\mu \\ (i,j)\in\mu,\ (j,\ell)\in\mu'}} f'_{j\ell} \circ f_{ij} \oplus \bigoplus_{i\in I\setminus\pi_I(\mu)} g_i \oplus \bigoplus_{\ell\in L\setminus\pi_L(\mu')} h'_\ell,$$

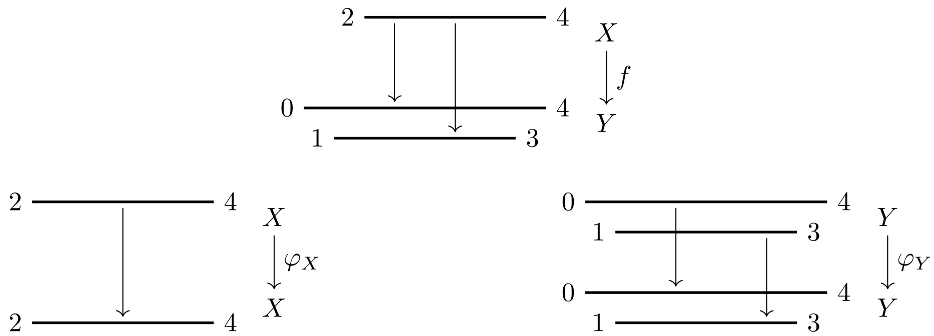


Figure 2 Example of a non bar-to-bar morphism $f : X \to Y$. The only valid automorphisms $\varphi_X : X \to X$, $\varphi_Y : Y \to Y$ are represented below. In both cases, the image of the restriction of the automorphism to a bar is contained within that same bar. Consequently, they do not alter the morphism when composed.

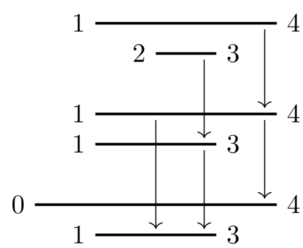


Figure 3 Representation of the morphisms in Bar of Remark 3.3.

where $\mu' \circ \mu$ is the set-theoretic matching that results from composing μ and μ' as multiset matchings, and the j in the first sum is unique, given $(i, \ell) \in \mu' \circ \mu$. We observe that $f'_{j\ell} \circ f_{ij}$ is nonzero if, and only if, (i, ℓ) is in the overlap composition $\mu' \bullet \mu$ (see Remark 2.4). If not, then $i \in I \setminus \pi_I(\mu' \bullet \mu)$, $\ell \in L \setminus \pi_L(\mu' \bullet \mu)$, and $f'_{j\ell} \circ f_{ij}$ is equal to the sum of zero maps $(K(a_i, b_i) \to 0) \oplus (0 \to K(e_\ell, f_\ell))$. Thus we can write

$$f' \circ f = \bigoplus_{\substack{(i,\ell)\in\mu'\bullet\mu \\ (i,j)\in\mu,\ (j,\ell)\in\mu'}} f'_{j\ell} \circ f_{ij} \oplus \bigoplus_{i\in I\setminus\pi_I(\mu'\bullet\mu)} g_i \oplus \bigoplus_{\ell\in L\setminus\pi_L(\mu'\bullet\mu)} h'_\ell,$$

which is an equation of the form (3), and this concludes the proof. ◀

▶ Remark 3.3. The composition of arbitrary bar-to-bar monomorphisms is not necessarily bar-to-bar. For example, consider the following composition of morphisms (see Figure 3):

$$K(1,4) \oplus K(2,3) \xrightarrow{\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}} K(1,4) \oplus K(1,3) \xrightarrow{\begin{bmatrix}1 & 0\\ 1 & 1\end{bmatrix}} K(0,4) \oplus K(1,3),$$

where the sign decorations of the start- and end-points are not important. Clearly the left map is bar-to-bar. The right map is bar-to-bar by doing a base change on the middle persistence module. However, the composition is a morphism between persistence modules with trivial automorphism groups, but its matrix representation looks like that of the right morphism, and therefore the composition morphism is not bar-to-bar.

We can view the matching μ of Definition 3.1 as a matching of barcodes. Specifically, μ induces the multiset matching (M, m_M) where $M = \mathcal{B}(X) \times \mathcal{B}(Y)$ and, for all $(\langle a, b\rangle, \langle c, d\rangle) \in$

M,

$$m_M(\langle a,b\rangle, \langle c,d\rangle) := \#\{(i,j) \in \mu \mid (a_i,b_i) = (a,b) \text{ and } (c_j,d_j) = (c,d)\}.$$

It turns out that this matching is uniquely determined as soon as we have a bar-to-bar morphism in Pfd:

▶ **Proposition 3.4.** *Let $f\colon X \to Y$ be a bar-to-bar morphism in* Pfd*, and let:*

- *$\varphi_X : X \cong \bigoplus_{i\in I} K(a_i,b_i)$ and $\varphi_Y : Y \cong \bigoplus_{j\in J} K(c_j,d_j)$ be barcode decompositions;*
- *$g\colon \bigoplus_{i\in I} K(a_i,b_i) \to \bigoplus_{j\in J} K(c_j,d_j)$ be a strictly bar-to-bar morphism in* Bar *isomorphic to f (which exists by definition);*
- *μ be a matching in* $\mathsf{Match}(I,J)$ *associated to g (see Definition 3.1).*

Then the multiset matching $M \in \mathsf{Match}_0(\mathcal{B}(X), \mathcal{B}(Y))$ induced by μ does not depend on φ_X, φ_Y, g, nor μ.

Proof. We view f as an object in the category $\mathsf{Fun}(\mathbb{R} \times \{0 < 1\}, \mathrm{vect}_K)$. As such a representation, the decomposition (3) of g becomes

$$f \cong g \cong \bigoplus_{(i,j)\in\mu} \mathbf{R}^{\langle a_i,b_i\rangle}_{\langle c_j,d_j\rangle} \oplus \bigoplus_{i\in I\setminus\pi_I(\mu)} \mathbf{I}^{\langle a_i,b_i\rangle}_0 \oplus \bigoplus_{j\in J\setminus\pi_J(\mu)} \mathbf{I}^0_{\langle c_j,d_j\rangle},$$

where the indecomposables $\mathbf{R}^*_*$ and $\mathbf{I}^*_*$ are adapted from [12, Definition 5.2] and defined as follows:

- $\mathbf{R}^{\langle a_i,b_i\rangle}_{\langle c_j,d_j\rangle}$, where $\langle a_i,b_i\rangle$ overlaps $\langle c_j,d_j\rangle$ above, is the representation of a nonzero morphism $K(a_i,b_i) \to K(c_j,d_j)$.
- $\mathbf{I}^{\langle a_i,b_i\rangle}_0$ is the representation of the zero morphism $K(a_i,b_i) \to 0$.
- Similarly, $\mathbf{I}^0_{\langle c_j,d_j\rangle}$ is the representation of the zero morphism $0 \to K(c_j,d_j)$.

By the Krull-Remak-Schimdt-Azumaya theorem [2] (see also [5]), this decomposition is unique up to permutation of the indecomposables. This implies that the matching μ is unique up to permutations of indices i, i' in I such that $a_i = a_{i'}$ and $b_i = b_{i'}$, and similarly for J. In particular, the induced matching M on barcodes is unique. The fact that M is an overlap matching is clear from its construction, and this concludes the proof. ◀

3.2 The same-endpoint operator

We define the same-endpoint operator on direct sums of bars, and then show that it is well-defined up to isomorphism on arbitrary persistence modules in Pfd.

▶ **Definition 3.5** (same-endpoint operator Δ)**.** *We define Δ as the operator* Bar $\to$ Bar *that is the identity on objects and the projection*

$$\hom(X,Y) \cong \bigoplus_{i,j} \hom(K(a_i,b_i), K(c_j,d_j)) \longrightarrow \bigoplus_{\substack{i,j\\ b_i = d_j}} \hom(K(a_i,b_i), K(c_j,d_j)) \hookrightarrow \hom(X,Y)$$

*on morphisms, where $X = \bigoplus_i K(a_i,b_i)$ and $Y = \bigoplus_j K(c_j,d_j)$. We call Δ the **same-endpoint operator**.*

▶ Remark 3.6. The first isomorphism and the last inclusion are canonical. Thus the operator Δ is well-defined and does not depend on any arbitrary choices.

▶ **Example 3.7.** Figure 4 shows an example of applying the same-endpoint operator to a monomorphism in Bar.

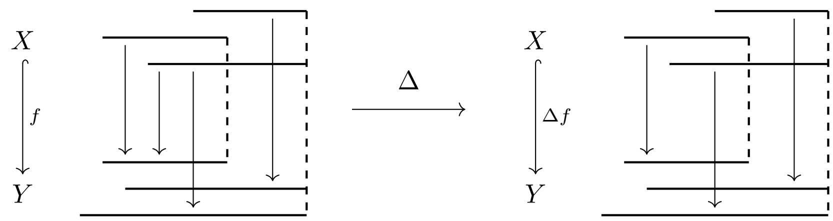


Figure 4 same-endpoint operator applied to a monomorphism $f\colon X \hookrightarrow Y$ in Bar. The resulting monomorphism Δf is obtained by setting to zero all the nonzero projections between bars with different end points.

▶ **Proposition 3.8.** *The operator Δ is linear on hom-spaces.*

Proof. Immediate from its definition as a linear projection on hom-spaces. ◀

▶ **Proposition 3.9.** *The operator Δ is a functor.*

Proof. Clearly $\Delta\,\mathrm{id} = \mathrm{id}$. For compatibility with composition, consider $f\colon \bigoplus_i K(a_i,b_i) \to \bigoplus_j K(c_j,d_j)$ and $g\colon \bigoplus_j K(c_j,d_j) \to \bigoplus_k K(s_k,t_k)$ composable morphisms. We write f_{ij} for the corresponding morphism

$$\bigoplus_i K(a_i,b_i) \xrightarrow{p_i} K(a_i,b_i) \to K(c_j,d_j) \xrightarrow{\iota_j} \bigoplus_j K(c_j,d_j)$$

and similarly g_{jk} and $(g\circ f)_{ik}$. Then

$$g\circ f = \sum_{i,j,k} g_{jk}\circ f_{ij} = \sum_{\substack{i,j,k\\ b_i\ge d_j\ge t_k}} g_{jk}\circ f_{ij} \quad \text{and} \quad (g\circ f)_{ik} = \sum_{\substack{j\\ b_i\ge d_j\ge t_k}} g_{jk}\circ f_{ij}$$

because all other summands are 0. Moreover,

$$\Delta f = \sum_{b_i=d_j} f_{ij}, \quad \Delta g = \sum_{d_j=t_k} g_{jk}, \quad \text{and} \quad \Delta(g\circ f) = \sum_{b_i=t_k} (g\circ f)_{ik}.$$

Thus we get

$$(\Delta(g\circ f))_{ik} = \sum_{\substack{j\\ b_i\ge d_j\ge t_k\\ b_i=t_k}} g_{jk}\circ f_{ij} = \sum_{\substack{j\\ b_i=d_j=t_k}} g_{jk}\circ f_{ij} = (\Delta g\circ\Delta f)_{ik},$$

which shows the compatibility with composition. ◀

▶ **Corollary 3.10.** *Let $f\colon X\hookrightarrow Y$ be a monomorphism in* Bar. *Then*

1. *If f is bar-to-bar, then it is isomorphic to Δf.*
2. *If f is strictly bar-to-bar, then $f = \Delta f$.*

Saying that two morphisms $f, g\colon X \to Y$ are **isomorphic** means that there exist automorphisms $\varphi\colon X\to X$ and $\psi\colon Y\to Y$ such that $f = \psi\circ g\circ\varphi$.

Proof. If f is a strictly bar-to-bar monomorphism, then it is of the form

$$\bigoplus_i (K(a_i,b_i) \hookrightarrow K(c_i,b_i)) \oplus \bigoplus_j (0 \to K(c_j,d_j))$$

where $X = \bigoplus_i K(a_i, b_i)$ and $Y = \bigoplus_i K(c_i, b_i) \oplus \bigoplus_j K(c_j, d_j)$. The second statement then follows from the definition of Δ.

For the first statement, by definition, there exist automorphisms $\varphi_X : X \to X$ and $\varphi_Y : Y \to Y$ such that $\varphi_Y \circ f \circ \varphi_X^{-1}$ is strictly bar-to-bar. Then

$$\varphi_Y \circ f \circ \varphi_X^{-1} = \Delta(\varphi_Y \circ f \circ \varphi_X^{-1}) = \Delta\varphi_Y \circ \Delta f \circ (\Delta\varphi_X)^{-1}.$$

Since Δ is a functor, it sends isomorphisms to isomorphisms. Thus Δf is isomorphic to f. ◀

▶ Remark 3.11. The converse of the first statement is also true, but it follows from Theorem 3.15. The converse of the second statement is not true: see the right side of **??**, where $f = \Delta f$ but f is not strictly bar-to-bar.

▶ **Proposition 3.12.** *Let $f : X \hookrightarrow Y$ be a monomorphism in* Bar*. Then Δf is also a monomorphism.*

Proof. Write $X = \bigoplus_i K(a_i, b_i)$ and $Y = \bigoplus_j K(c_j, d_j)$. Recall that Δf is a direct sum of morphisms $\bigoplus_{b_i = b} K(a_i, b_i) \to \bigoplus_{d_j = b} K(c_j, d_j)$ for all $b \in \mathbb{E}$. Thus it suffices to prove, for all $b \in \mathbb{E}$, that Δf restricted to $X^{(b)} \coloneqq \bigoplus_{b_i = b} K(a_i, b_i)$ is a monomorphism.

Fix $b \in \mathbb{E}$ and write $g \coloneqq \Delta f|_{X^{(b)}}$. There exists $t \in \mathbb{R}$ such that $t^- < b$ and every bar $K(a_i, b)$ in X satisfies $a_i \le t^-$. If there were an infinite number of bars $K(c_j, d_j)$ in Y such that $t \in \langle c_j, d_j \rangle$, then Y_t would be infinite-dimensional. Since Y is in Pfd, this is not possible, so we conclude that there are only finitely many such bars, and therefore there exists $u \ge t$ in $\mathbb{R}$ such that f maps bars $K(a_i, b_i)$ in X with $b_i = b$ exclusively to bars $K(c_j, d_j)$ in Y with $d_j = b$. Thus, at u, the same-endpoint operator does not affect $f|_{X^{(b)}}$: in other words, $g_u = (f|_{X^{(b)}})_u$. In particular, $g_u = (f|_{X^{(b)}})_u$ is injective.

Next, we check that g_s is injective for all $s < u$. Indeed, if g_s sent a nonzero element $x \in X_s^{(b)}$ to $0 \in Y_s$, then by the transition maps we would get $X_{s \le u}(x) \neq 0$ and $g_t(X_{s \le u}(x)) = 0$, contradicting the injectivity of g_u. For $s \in \langle u^+, b \rangle$, injectivity holds because $g_s = (f|_{X^{(b)}})_s$ by the same argument as before. Thus g is pointwise injective, so it is a monomorphism. This concludes the proof. ◀

▶ **Proposition 3.13.** *Let $f : X \to Y$ be a morphism in* Pfd*. Let $\varphi_X : X \cong \bigoplus_i K(a_i, b_i)$ and $\varphi_Y : Y \cong \bigoplus_j K(c_j, d_j)$ be barcode decompositions of X and Y. Define $\Delta f \coloneqq \varphi_Y^{-1} \Delta(\varphi_Y f \varphi_X^{-1}) \varphi_X$, where we identify the morphism $\varphi_Y f \varphi_X^{-1}$ in* Pfd *with the one in* Bar*. Then, up to isomorphism, Δf does not depend on the choice of φ_X and φ_Y.*

Proof. Let ψ_X and ψ_Y be other choices of isomorphisms. Then

$$\begin{aligned}
\psi_Y^{-1} \Delta(\psi_Y f \psi_X^{-1}) \psi_X &= \psi_Y^{-1} \Delta(\psi_Y \varphi_Y^{-1} \varphi_Y f \varphi_X^{-1} \varphi_X \psi_X^{-1}) \psi_X \\
&\cong \Delta(\underbrace{\psi_Y \varphi_Y^{-1}}_{\in \mathsf{Bar}} \underbrace{\varphi_Y f \varphi_X^{-1}}_{\in \mathsf{Bar}} \underbrace{\varphi_X \psi_X^{-1}}_{\in \mathsf{Bar}}) \\
&= \Delta(\psi_Y \varphi_Y^{-1}) \Delta(\varphi_Y f \varphi_X^{-1}) \Delta(\varphi_X \psi_X^{-1}) \qquad \text{(by functoriality in Bar)} \\
&\cong \Delta(\varphi_Y f \varphi_X^{-1}) \\
&\cong \varphi_Y^{-1} \Delta(\varphi_Y f \varphi_X^{-1}) \varphi_X.
\end{aligned}$$

◀

▶ Remark 3.14. In particular, this means that we can define the same-endpoint operator Δ on Pfd up to isomorphism.

Algorithm 1 Infinite column-finite matrix reduction. Matrix indices start from 1, I denotes the infinite identity matrix, $E_{i,j}$ denotes the infinite matrix with all 0's except a 1 in position (i, j), and $\mathrm{low}(j) \coloneqq \max\{i \geq 1 \mid A_{i,j} \neq 0\}$.

Input: an infinite column-finite matrix M.
Output: a factorization $A = UMV$ where U and V are upper triangular invertible and A has at most one nonzero entry in each row and column.

```
U ← I;
V ← I;
A ← M;
for j = 1, 2, ... do
    while ∃j' < j, low(j') = low(j) do      // reduce by columns with same low
        λ ← A_{low(j),j} / A_{low(j'),j'};
        A_{*,j} ← A_{*,j} − λA_{*,j'} = A(I − λE_{j',j});
        V ← V(I − λE_{j',j});
    end
    i ← low(j);
    while ∃i' < i, A_{i',j} ≠ 0 do          // reduce nonzero entries above low
        λ ← A_{i',j} / A_{i,j};
        A_{i',*} ← A_{i',*} − λA_{i,*} = (I − λE_{i',i})A;
        U ← (I − λE_{i',i})U;
    end
end
return A, U, V
```

▶ **Theorem 3.15.** *For all morphisms f in* Pfd*, the morphism Δf is bar-to-bar.*

▶ Remark 3.16. In Pfd, the morphism Δf is defined only up to isomorphism, see Proposition 3.13. Since the property of being bar-to-bar is an invariant of isomorphism classes of morphisms (by definition), Theorem 3.15 is well-formulated.

To show Theorem 3.15, we will be working with infinite matrices. In particular with infinite column-finite matrices (see Definition 2.13).

In [13] the author presents a reduction algorithm for infinite row-finite matrices (defined analogously to column-finite ones). We proceed similarly with Algorithm 1: compare with the standard algorithm of persistent homology, see [8].

▶ **Lemma 3.17.** *Algorithm 1 is well-specified: it takes as input an infinite column-finite matrix M and returns a factorization $A = UMV$ where U and V are upper triangular invertible and A has at most one nonzero entry in each row and column.*

Proof. We observe that after the j^{th} step, the first j columns of V are fixed (they will not be modified in subsequent iterations). To show that the limit matrix V is invertible, we construct its left inverse iteratively, starting with $V^{-1} = I$ and then applying $V^{-1} \leftarrow (I - \lambda E_{j',j})V^{-1}$ every time we set $V \leftarrow V(I - \lambda E_{j',j})$. After the j^{th} step, the row transformations on V^{-1} involve adding rows with index $k > j$ to other rows: this will never affect the first j rows, as V^{-1} is always upper triangular.

For U, the argument is a little subtler. For all j, there exists a positive integer J such that, in the fully reduced matrix A, for all $j' > J$, $\mathrm{low}(j') > \mathrm{low}(j)$. Thus, after the J^{th} step, the j^{th} column of U is no longer affected by the row transformations that update U. To show that the limit matrix U is invertible, we construct its right inverse iteratively, starting with $U^{-1} = I$ and then applying $U^{-1} \leftarrow U^{-1}(I - \lambda E_{i',i})$ every time we set $U \leftarrow (I - \lambda E_{i',i})U$. After the J^{th} step, the only columns that get transformed are those with index $> \text{low}(j)$, so in particular the j^{th} column is no longer affected by the column transformations that update U^{-1}.

Finally, for A, we use the same arguments as for V and U to conclude that A converges to a well-defined limit matrix. ◀

Proof of Theorem 3.15. It suffices to prove the result for f in Bar, so we assume this. In this case, the morphism $\Delta f\colon X \to Y$ is a direct sum of morphisms between the direct summands of X and Y where all bars share a common end-point. Δf is bar-to-bar if each summand is bar-to-bar, so, without loss of generality, we can assume that X and Y are persistence modules where all bars share a common end-point. Following Remark 2.8, we can then represent Δf as an infinite matrix M with countably many columns and rows, and where the columns and rows correspond to bars and are sorted by start-point.

We observe that every column of M has a finite number of nonzero entries. Indeed, if the column corresponding to a bar $K(a, b)$ of X had an infinite number of nonzero entries, then $K(a, b)$ would map to an infinite sequence $(K(c_i, b))_i$ of bars of Y, with $c_i \leq a$ for all i. Then (since we do not consider empty bars), there exists a point $t \in \langle a, b\rangle \subseteq \langle c_i, b\rangle$ for all i, and so Y_t is infinite-dimensional, which contradicts the definition of Bar.

Thus we can apply Algorithm 1 to obtain invertible upper triangular matrices U and V such that UMV has at most one nonzero entry in each row and column: this corresponds to a strictly bar-to-bar morphism from X to Y. Since U and V are upper triangular, they correspond to automorphisms of X and Y, respectively. Therefore the morphism f is isomorphic to a strictly bar-to-bar morphism. In other words, f is bar-to-bar. ◀

3.3 Induced matchings

We use the definitions of **multiset representations**, **barcodes**, and the category **Barc** from [4]. While for their canonically induced matchings compose naturally, we need more structure on our persistence modules to make the matchings compose in an intuitive way.

▶ **Definition 3.18** (Matching operators). *A* ***matching operator*** *is a function μ that takes a morphism $f\colon X \to Y$ in Bar and returns a matching (i.e., morphism) $\mu(f)\colon \mathcal{B}(X) \to \mathcal{B}(Y)$ in* **Barc**. *The second argument is sometimes omitted. The operator μ is*

1. ***additive*** *if, given morphisms f and g, $\mu(f \oplus g) = \mu(f) \sqcup \mu(g)$.*
2. ***split*** *if, for a strictly bar-to-bar morphism f, $\mu(f)$ is the matching of Definition 3.1*[1]*.*
3. ***functorial on monomorphisms*** *(resp.* ***epimorphisms****) if, given composable monomorphisms (resp. epimorphisms) f and g, $\mu(g \circ f) = \mu(g) \bullet \mu(f)$.*

▶ Remark 3.19. **1.** Matching operators are well-defined. In particular, they are equivariant under reindexing of the barcode multiset representations [4].

2. By [3, Proposition 5.10], such operators are never functorial over all morphisms. Moreover, the **canonically induced matching** of that paper is functorial on monomorphisms and epimorphisms.
3. Note that being split is equivalent to being additive and mapping morphisms $K(a, b) \to K(c, d)$ between single bars nontrivially.

[1] more precisely, the definition's matching is $\{(i, j) \mid ((\langle a_i, b_i\rangle, i), (\langle c_j, d_j\rangle, j)) \in \mu(f)\}$.

4. Using the axiom of choice, we can define arbitrary additive matchings by choosing a matching for each indecomposable morphism.

We now show that being split and functorial on monomorphisms (resp. epipmorphisms) are mutually exclusive.

▶ **Proposition 3.20.** *A matching operator cannot be split and functorial on monomorphisms (resp. epimorphisms).*

Proof. We proceed by contradiction: suppose that such an operator μ exists. Let $b > a_1 > a_2 > \cdots > a_6$ be elements of $\mathbb{E}$, and consider the sequence

$$X = K(a_1, b) \oplus K(a_2, b) \xhookrightarrow{f} Y = K(a_3, b) \oplus K(a_4, b) \xhookrightarrow{g} Z = K(a_5, b) \oplus K(a_6, b)$$

where the two monomorphisms are represented by the labeled matrices

$$f = \begin{array}{cc} & \begin{array}{cc} a_2 & a_1 \end{array} \\ \begin{array}{c} a_4 \\ a_3 \end{array} & \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \end{array}, \quad g = \begin{array}{cc} & \begin{array}{cc} a_4 & a_3 \end{array} \\ \begin{array}{c} a_6 \\ a_5 \end{array} & \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \end{array}, \quad \text{and so} \quad gf = \begin{array}{cc} & \begin{array}{cc} a_2 & a_1 \end{array} \\ \begin{array}{c} a_6 \\ a_5 \end{array} & \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \end{array},$$

where the label a_i corresponds to the interval $\langle a_i, b\rangle$. Recall that isomorphisms of morphisms correspond to certain left-to-right column transformations and bottom-to-top row transformations of their matrix representations. Adding the left column to the right column in the matrix representation of gf, we obtain the isomorphism

$$gf \cong \begin{array}{cc} & \begin{array}{cc} a_2 & a_1 \end{array} \\ \begin{array}{c} a_6 \\ a_5 \end{array} & \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \end{array} = \underbrace{\begin{array}{cc} & \begin{array}{cc} a_4 & a_3 \end{array} \\ \begin{array}{c} a_6 \\ a_5 \end{array} & \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \end{array}}_{=:\, h_0} \circ \begin{array}{cc} & \begin{array}{cc} a_2 & a_1 \end{array} \\ \begin{array}{c} a_4 \\ a_3 \end{array} & \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \end{array}. \tag{4}$$

In other words, gf and $h_0 f$ are isomorphic. Moreover, we can decompose f as

$$\begin{array}{cc} & \begin{array}{cc} a_2 & a_1 \end{array} \\ \begin{array}{c} a_4 \\ a_3 \end{array} & \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \end{array} = \underbrace{\begin{array}{cc} & \begin{array}{cc} a_4 & a_3 \end{array} \\ \begin{array}{c} a_4 \\ a_3 \end{array} & \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \end{array}}_{=:\, h_1} \circ \underbrace{\begin{array}{cc} & \begin{array}{cc} a_2 & a_1 \end{array} \\ \begin{array}{c} a_4 \\ a_3 \end{array} & \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \end{array}}_{=:\, h_2}$$

The morphism h_1 is an isomorphism, and by splitness $\mu(h_2)$ is also an isomorphism, so by functoriality (recall also that, by functoriality, μ sends isomorphisms to isomorphisms), we deduce that $\mu(f) = \mu(h_1) \circ \mu(h_2)$ is an isomorphism. Applying μ to (4), by functoriality, we get $\mu(g) \circ \mu(f) \cong \mu(h_0) \circ \mu(f)$. As we just showed that $\mu(f)$ is invertible, we deduce that $\mu(g) \cong \mu(h_0)$. But the only automorphisms of $\mathcal{B}(Y)$ and $\mathcal{B}(Z)$ are the identity matchings, so $\mu(g) = \mu(h_0)$. But this contradicts the assumption that μ is split, since g and h_0 are different strictly bar-to-bar morphisms. We conclude that there is no induced matching representation operator that is both split and functorial on monomorphisms. The epimorphism case holds by a dual argument. ◀

We can now define our induced matching. Given a morphism $f\colon X \to Y$ in Bar, we know by Theorem 3.15 that Δf is bar-to-bar. Specifically, Algorithm 1 induces an overlap matching $\mu_\Delta(f) \in \mathbf{Barc}(\mathcal{B}(X), \mathcal{B}(Y))$. However, this matching is not very informative. Indeed, it is empty as soon as no bars of X or Y have the same end-point. Instead, we may follow the strategy of [3]: given a morphism $f\colon X \to Y$ in Bar, consider an epi-mono factorization of f,

$$X \xrightarrow{e} \mathrm{im}(f) \xrightarrow{m} Y,$$

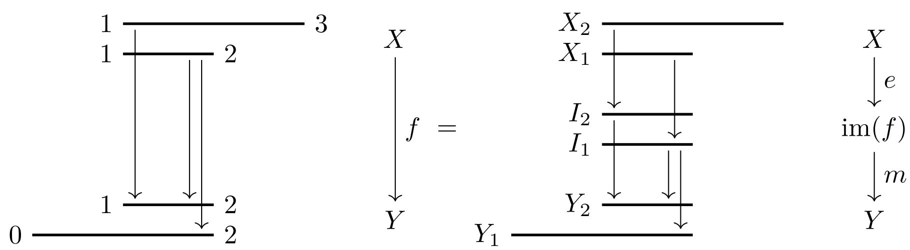
where we define $\mathrm{im}(f)$ as the object in Bar that isomorphic to the image of f in Pfd, i.e. it is the sum of the bars in the barcode of the image. We would then define the **matching induced by** f to be the matching

$$\mu(f) := \mu_\Delta(m) \bullet \mu_\Delta(e^*)^* \tag{5}$$

in **Barc**, where $-^*$ is the dual operator for epimorphisms and matchings (see Section 2.4).

The issue with this construction is that the epi-mono factorization is only unique up to isomorphism: there exist other factorizations, which are all of the form $(\varphi^{-1} \circ e, m \circ \varphi)$ with φ an automorphism on $\mathrm{im}(f)$. Indeed, we have the following example, which shows that the induced matching depends on the choice of factorization.

▶ **Example 3.21.** Consider the following morphism (over the field $\mathbb{F}_2$) along with an epi-mono factorization:



where, on the right, we have named all of the bars. The image $\mathrm{im}(f)$ consists of two bars I_1 and I_2 with identical support $\langle 1, 2\rangle$. We have the matrix representations

$$F = \begin{array}{c c} & \begin{array}{cc} X_1 & X_2 \end{array} \\ \begin{array}{c} Y_1 \\ Y_2 \end{array} & \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \end{array}, \quad E = \begin{array}{c c} & \begin{array}{cc} X_1 & X_2 \end{array} \\ \begin{array}{c} I_1 \\ I_2 \end{array} & \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \end{array},$$

$$\text{and} \quad M = \begin{array}{c c} & \begin{array}{cc} I_1 & I_2 \end{array} \\ \begin{array}{c} Y_1 \\ Y_2 \end{array} & \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \end{array} \cong \begin{array}{c c} & \begin{array}{cc} I_1 & I_2 \end{array} \\ \begin{array}{c} Y_1 \\ Y_2 \end{array} & \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \end{array} \begin{array}{c c} & \begin{array}{cc} I_1 & I_2 \end{array} \\ \begin{array}{c} I_1 \\ I_2 \end{array} & \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \end{array} = \begin{array}{c c} & \begin{array}{cc} I_1 & I_2 \end{array} \\ \begin{array}{c} Y_1 \\ Y_2 \end{array} & \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \end{array}$$

for f, e, and m, respectively (recall that we are working over the field $\mathbb{F}_2$). The induced matching is

$$\begin{aligned} \mu_\Delta(m) \bullet \mu_\Delta(e^*)^* &= \{(I_1, Y_2), (I_2, Y_1)\} \bullet \{(X_1, I_1), (X_2, I_2)\} \\ &= \{(X_1, Y_2), (X_2, Y_1)\}. \end{aligned}$$

Now consider $\varphi\colon \mathrm{im}(f) \to \mathrm{im}(f)$ the automorphism that permutes the two bars, with matrix representation

$$\Phi = \begin{array}{c c} & \begin{array}{cc} I_1 & I_2 \end{array} \\ \begin{array}{c} I_1 \\ I_2 \end{array} & \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \end{array}.$$

Since $\Phi^{-1} = \Phi$, we compute

$$\Phi^{-1}E = \begin{array}{c c} & \begin{array}{cc} X_1 & X_2 \end{array} \\ \begin{array}{c} I_1 \\ I_2 \end{array} & \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \end{array} \quad \text{and} \quad M\Phi = \begin{array}{c c} & \begin{array}{cc} I_1 & I_2 \end{array} \\ \begin{array}{c} Y_1 \\ Y_2 \end{array} & \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \end{array} \cong \begin{array}{c c} & \begin{array}{cc} I_1 & I_2 \end{array} \\ \begin{array}{c} Y_1 \\ Y_2 \end{array} & \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \end{array}$$

and find the induced matching

$$\begin{aligned} \mu_\Delta(m \circ \varphi) \bullet \mu_\Delta((\varphi^{-1} \circ e)^*)^* &= \{(I_1, Y_2), (I_2, Y_1)\} \bullet \{(X_1, I_2), (X_2, I_1)\} \\ &= \{(X_1, Y_1), (X_2, Y_2)\} \\ &\neq \mu_\Delta(m) \bullet \mu_\Delta(e^*)^*. \end{aligned}$$

Thus the induced matching as defined in Equation (5) is not unique.

▶ **Lemma 3.22.** *In the category* Bar*, let $f\colon X \to Y$ be a morphism and $\varphi\colon X \to X$ and $\psi\colon Y \to Y$ automorphisms. Then there exist automorphic matchings $\sigma\colon \mathcal{B}(X) \to \mathcal{B}(X)$ and $\tau\colon \mathcal{B}(Y) \to \mathcal{B}(Y)$ such that*

$$\mu_\Delta(\psi \circ f \circ \varphi) = \tau \bullet \mu_\Delta(f) \bullet \sigma.$$

Proof. The morphisms f and $\psi \circ f \circ \varphi$ are isomorphic to each other. By Proposition 3.13, $\Delta(f)$ and $\Delta(\psi \circ f \circ \varphi)$ are isomorphic. By definition of μ_Δ and the arguments of the proof of Proposition 3.4 (recall that the induced matching is read from the decomposition of bar-to-bar morphisms viewed as ladder representations), the induced matchings $\mu_\Delta(f)$ and $\mu_\Delta(\psi \circ f \circ \varphi)$ are equal up to permutations of bars with identical support. By the overlap condition for composition in **Barc**, automorphic matchings are exactly the bijective matchings that, at most, only permute of bars with identical support. We conclude that there exist automorphic matchings $\sigma\colon \mathcal{B}(X) \to \mathcal{B}(X)$ and $\tau\colon \mathcal{B}(Y) \to \mathcal{B}(Y)$ such that

$$\mu_\Delta(\psi \circ f \circ \varphi) = \tau \bullet \mu_\Delta(f) \bullet \sigma. \qquad ◀$$

▶ **Example 3.23.** The automorphic matchings depend on the morphism f. Consider the following example. Let f and $g\colon X \to Y$ be morphisms in Bar represented by the matrices

$$F = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \cong \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \quad \text{and} \quad G = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix},$$

respectively, where the two bars of X have the same support, and the two bars of Y have the same support (but not necessarily the same as that of X). Let φ be the automorphism on X represented by

$$\Phi = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Then

$$F\Phi = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \cong \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \text{and} \quad G\Phi = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Reading the pivots of the relevant matrices, we find that $\mu_\Delta(f) = \mu_\Delta(f\varphi)$ but $\mu_\Delta(g) \neq \mu_\Delta(g\varphi)$. In other words, the automorphic matchings are not the same (phrase this better).

▶ **Example 3.24.** We can define an induced matching on Pfd as soon as we determine a strategy for selecting, for a given morphism f in Pfd, an epi-mono factorization $f = me$. As an example, we recover the induced matchings of [10, 11] via epi-mono factorizations.

Let $f\colon X \to Y$ be a morphism in Bar. Denote by F the matrix representation of f where rows are ordered (lexicographically) by end-point then start-point and the columns by start-point then end-point. For each column in F, reduce it by columns to its left. This is a finite process because F is column-finite. This gives us $R = FV$ with R a matrix with unique pivots and V upper triangular and invertible.

Suppose that f is tame (needed to apply the results that give the image barcode), and write $X = \bigoplus_{j=1}^{n} K(a_j, b_j)$ and $Y = \bigoplus_{i=1}^{m} K(c_i, d_i)$. Note the inversion of indices, chosen for compatibility with matrix index notation. By [11, Theorem 5.1] (see also [16, Proposition 3.17]), the barcode of $\mathrm{im}(f)$ consists of, for each pivot (i, j) in R whose column is labeled by $\langle a_j, b_j\rangle$ and whose row is labeled by $\langle c_j, d_i\rangle$, a bar of support $\langle a_j, d_i\rangle$.

Let I be the subset of indices in $\{1, \dots, m\}$ such that row i of R contains a pivot. Let E be a matrix whose columns are labeled like F and whose rows are indexed by I. We label row i of E by the interval $\langle a_j, d_i\rangle$, corresponding to the pivot (i, j) of R. We set

$$E_{ij} \coloneqq \begin{cases} R_{ij} & \text{if } i = \mathrm{low}(j) \text{ in } R, \\ 0 & \text{otherwise.} \end{cases}$$

We see that E represents an epimorphism from X to $\mathrm{im}(f)$.

Let M be a matrix that solves $ME = R$ (this determines the labeling of M). Let us show that this solution exists. By construction, E has at most one nonzero coefficient in each column. Therefore, a given unknown $M_{i,i'}$ in M appears in at most one equation:

- If there exists $j \in \{1, \dots, n\}$ such that $i' = \mathrm{low}(j)$ in R, then $E_{i',j}$ is nonzero and $M_{i,i'} = M_{i,i'}$ must satisfy the equation

  $$M_{i,i'} E_{i',j} = R_{ij}.$$

  If $R_{ij} = 0$, then we set $M_{i,i'} = 0$. Otherwise, we need to check that we can set $M_{i,i'}$ to be nonzero. We know that $i \leq i'$ ($R_{i',j}$ is the pivot, so R_{ij} must be above it). Column i' of M is labeled by $\langle a_j, d_{i'}\rangle$ and row i of M is labeled by $\langle c_i, d_i\rangle$. We have $d_{i'} \geq d_i$ because $i' \geq i$ and we assume that the $\langle c_i, d_i\rangle$ are ordered compatibly with the product order. We also have $a_j \geq c_i$ because R_{ij} is nonzero (and so $\langle a_j, b_j\rangle$ must overlap $\langle c_i, d_i\rangle$ above). Thus we have checked that $\langle a_j, d_{i'}\rangle$ overlaps $\langle c_i, d_i\rangle$ above, so we can set $M_{i,i'}$ to be nonzero.
- Otherwise, $M_{i,i'}$ has no equations to satisfy, and we choose to set $M_{i,i'} = 0$.

We observe that the resulting M is in reduced row echelon form, with pivots at the coordinates (i, i) for i in I, and it represents a monomorphism (need some details here).

We thus obtain $F = MEV^{-1}$, giving us an epi-mono factorization $f = me$ by the morphisms e and m represented by EV^{-1} and M, respectively.

We define the induced matching $\mu(f) \coloneqq \mu_\Delta(m) \bullet \mu_\Delta(e^*)^*$. By the last observation on M, we deduce that the induced matching $\mu_\Delta(m)$ matches each bar $\langle a_j, d_i\rangle$ in the image to $\langle c_i, d_i\rangle$ in $\mathcal{B}(Y)$. In other words, we know the matching as soon as we know $\mu_\Delta(e^*)^*$, or as soon as we know the reduced matrix R (up to permutation of identical bars).

4 Application to Wasserstein distances

4.1 Sizes and ranks of (co)kernels

In this section we prove some results in the category Pfd. In particular, we highlight a connection between rank functions and p-norms, which yields new proofs of results in [1, Section 3].

▶ **Proposition 4.1.** *Let $f\colon X \hookrightarrow Y$ be a monomorphism in* Pfd *and consider Δf the functorially defined bar-to-bar morphism and f_φ the bar-to-bar monomorphism induced by the canonical matching between X and Y (this assumes that we fix barcodes). Then, for all $s \le t$ in $[0,\infty)$,*

$$\operatorname{rk}((\operatorname{coker} f)_{s\le t}) \ge \operatorname{rk}((\operatorname{coker} \Delta f)_{s\le t}) \ge \operatorname{rk}((\operatorname{coker} f_\varphi)_{s\le t}).$$

We first prove a series of lemmas.

▶ **Lemma 4.2.** *Let $f\colon X \hookrightarrow Y$ be an arbitrary monomorphism in* Pfd*. For all $s \le t$ in $[0,\infty)$,*

$$\operatorname{rk}((\operatorname{coker} f)_{s\le t}) = \operatorname{rk}(Y_{s\le t} + f_t) - \dim X_t,$$

where $Y_{s\le t} + f_t \colon Y_s \oplus X_t \to Y_t$.

Proof. Consider the following diagram of vector spaces:

$$\begin{tikzcd}
X_s \arrow[r, hook, "f_s"] \arrow[d, "X_{s\le t}"'] & Y_s \arrow[r, two heads] \arrow[d, "Y_{s\le t}"'] & (\operatorname{coker} f)_s \arrow[d, "(\operatorname{coker} f)_{s\le t}"] \\
X_t \arrow[r, hook, "f_t"'] & Y_t \arrow[r, two heads] \arrow[d, two heads] & (\operatorname{coker} f)_t \arrow[d, two heads] \\
& \operatorname{coker} Y_{s\le t} \arrow[r, two heads] & C
\end{tikzcd}$$

where C is the cokernel of the map $(\operatorname{coker} f)_{s\le t}$. We observe (by a diagram chase, for instance) that C is also the cokernel of the map $g := Y_{s\le t} + f_t \colon Y_s \oplus X_t \to Y_t$. Thus $\dim C = \dim Y_t - \operatorname{rk} g$, and we conclude the proof by observing that

$$\begin{aligned}
\operatorname{rk}((\operatorname{coker} f)_{s\le t}) &= \dim(\operatorname{coker} f)_t - \dim C \\
&= \dim Y_t - \dim X_t - \dim Y_t + \operatorname{rk} g \\
&= \operatorname{rk} g - \dim X_t.
\end{aligned}$$

◀

▶ **Lemma 4.3.** *Let $g\colon X \to Y$ be an arbitrary morphism in* Pfd*. Then, for all t in $[0,\infty)$,*

$$\operatorname{rk} g_t \ge \operatorname{rk}(\Delta g)_t.$$

Proof. Fix barcode decompositions of X and Y. Then we can represent g_t as a (finite) matrix, where the columns and rows correspond to bars of X and Y that are nonzero at t, and they are sorted such that the death time is increasing. The matrix M of g_t is then block upper triangular, where the blocks correspond to bars with the same death time. The matrix M_Δ of $(\Delta g)_t$ is the block diagonal part of this matrix.

Let us show that the rank of M_Δ is smaller than the rank of M. Indeed, for every block of M_Δ, select a maximal set of linearly independent columns. The corresponding columns in M are also linearly independent, and therefore the linear map sending each of the selected columns of M_Δ to the corresponding column in M is injective. This defines an injective map $\operatorname{im} M_\Delta \hookrightarrow \operatorname{im} M$. Taking the dimension of both sides, we get $\operatorname{rk}(\Delta g)_t \leq \operatorname{rk} g_t$. ◀

▶ **Lemma 4.4.** *Let $f\colon X \hookrightarrow Y$ be an arbitrary monomorphism in* Pfd*. For all $s \leq t$ in $[0,\infty)$,*

$$\operatorname{rk}(Y_{s\leq t} + f_t) \geq \operatorname{rk}(Y_{s\leq t} + (\Delta f)_t),$$

where $Y_{s\leq t} + f_t$ and $Y_{s\leq t} + (\Delta f)_t\colon Y_s \oplus X_t \to Y_t$.

Proof. Since we are concerned with ranks, which are invariant by isomorphisms of morphisms, we can assume without loss of generality that X, Y, and f are in Bar.

Consider $Y[-\delta]$, the persistence module defined as $Y[-\delta]_a \coloneqq Y_{a-\delta}$. Then there exists a natural morphism in Pfd $Y_{\bullet-\delta\leq\bullet}\colon Y[-\delta] \to Y$. The image of this morphism then has the same barcode as Y, except that every birth time has been increased by δ, with resulting bars with a birth time greater than the death time being removed. Denote by $\iota\colon \operatorname{im} Y_{\bullet-\delta\leq\bullet} \hookrightarrow Y$ the monomorphism through which $Y_{\bullet-\delta\leq\bullet}$ factors. Then $\operatorname{rk}(Y_{s\leq t} + f_t) = \operatorname{rk}(\iota_t + f_t)$. Moreover, ι is strictly bar-to-bar, so by Corollary 3.10 $\Delta\iota = \iota$. By linearity of Δ (Proposition 3.8), we deduce that $\Delta(\iota + f) = \iota + \Delta f$. We conclude as follows:

$$\begin{aligned}
\operatorname{rk}(Y_{s\leq t} + f_t) &= \operatorname{rk}(\iota + f)_t \\
&\geq \operatorname{rk}(\Delta(\iota + f))_t && \text{(by Lemma 4.3)} \\
&= \operatorname{rk}(\iota_t + (\Delta f)_t) \\
&= \operatorname{rk}(Y_{s\leq t} + (\Delta f)_t).
\end{aligned}$$

◀

▶ **Lemma 4.5.** *Suppose that all of the bars of X and Y have the same end-point. Then*

$$\operatorname{rk}((\operatorname{coker} f_\varphi)_{s\leq t}) = (\operatorname{rk} Y_{s\leq t} - \dim X_t)_+,$$

where $(-)_+$ is the positive part function.

Proof. Denote by b this common end-point. Since f_φ is induced by the canonical matching, we can assume without loss of generality that $X = \bigoplus_{i=1}^m K(a_i, b)$, $Y = \bigoplus_{i=1}^n K(c_i, b)$ with $m \leq n$ in $\mathbb{N} \cup \{\omega\}$, the a_i and c_i are ordered in nondecreasing order, and the canonical matching φ is the inclusion of $\{1, \ldots, m\}$ in $\{1, \ldots, n\}$. The fact that X and Y have at most countably many bars with end-point b is explained in Remark 2.8. Then

$$f_\varphi = \bigoplus_{i=1}^m (K(a_i, b) \to K(c_i, b)) \oplus \bigoplus_{j=m+1}^n (0 \to K(c_i, b)).$$

Therefore

$$\operatorname{coker} f_\varphi = \bigoplus_{i=1}^m K(c_i, a_i) \oplus \bigoplus_{j=m+1}^n K(c_i, b).$$

If $t^- \geq b$, then the result holds trivially. Now consider the case $t^- < b$. Since the a_i and c_i are in nondecreasing order, there exist integers k_a and k_c such that

$$\{i \in \{1, \ldots, m\} : a_i \leq t^-\} = \{1, \ldots, k_a\} \quad \text{and} \quad \{i \in \{1, \ldots, n\} : c_i \leq s^-\} = \{1, \ldots, k_c\}.$$

Therefore

$$\{i \in \{1,\ldots,n\} : c_i \leq s^-\} \setminus \{i \in \{1,\ldots,m\} : a_i \leq t^-\} = \begin{cases} \{k_a+1,\ldots,k_c\} & \text{if } k_a < k_c, \\ \varnothing & \text{otherwise,} \end{cases}$$

and we conclude that

$$\begin{aligned} \operatorname{rk}((\operatorname{coker} f_\varphi)_{s\leq t}) &= |\{i \in \{1,\ldots,m\} : c_i \leq s^-, t^- < a_i\} \sqcup \{j \in \{m+1,\ldots,n\} : c_i \leq s^-\}| \\ &= |\{i \in \{1,\ldots,n\} : c_i \leq s^-\} \setminus \{i \in \{1,\ldots,m\} : a_i \leq t^-\}| \\ &= (k_c - k_a)_+ \\ &= (\operatorname{rk} Y_{s\leq t} - \dim X_t)_+, \end{aligned}$$

where the second equality comes from the equality between the sets, and the fourth equality comes from the fact that $k_a = \dim X_t$ and $k_c = \operatorname{rk} Y_{s\leq t}$. ◀

Proof of Proposition 4.1. We show the first inequality as follows:

$$\begin{aligned} \operatorname{rk}((\operatorname{coker} f)_{s\leq t}) &= \operatorname{rk}(Y_{s\leq t} + f_t) - \dim X_t && \text{(by Lemma 4.2)} \\ &\geq \operatorname{rk}(Y_{s\leq t} + (\Delta f)_t) - \dim X_t && \text{(by Lemma 4.4)} \\ &= \operatorname{rk}((\operatorname{coker} \Delta f)_{s\leq t}). && \text{(by Lemma 4.2)} \end{aligned}$$

We show the second inequality as follows. Since Δf is bar-to-bar by Theorem 3.15, its cokernel is the direct sum of the cokernels of the maps restricted to bars with a common end-point. Thus, without loss of generality, we can assume that the bars of X and Y all have the same end-point. In this case, we get

$$\begin{aligned} \operatorname{rk}((\operatorname{coker} \Delta f)_{s\leq t}) &= \operatorname{rk}(Y_{s\leq t} + (\Delta f)_t) - \dim X_t && \text{(by Lemma 4.2)} \\ &\geq \max\{\operatorname{rk} Y_{s\leq t}, \operatorname{rk}(\Delta f)_t\} - \dim X_t && \text{(true for general sums of maps)} \\ &= \max\{\operatorname{rk} Y_{s\leq t}, \dim X_t\} - \dim X_t && \text{(since } \Delta f \text{ is a monomorphism)} \\ &= (\operatorname{rk} Y_{s\leq t} - \dim X_t)_+ \\ &= \operatorname{rk}((\operatorname{coker} f_\varphi)_{s\leq t}). && \text{(by Lemma 4.5)} \end{aligned}$$

◀

To connect rank functions to p-norms of cokernels, we define the following measurable space.

▶ **Definition 4.6.** *Fix $p \in (1,\infty)$. We define the space $\Omega := \{(s,t) : s \leq t \in \mathbb{R}\}$ and the density function*

$$g_p(s,t) := p(p-1)|t-s|^{p-2}.$$

▶ **Proposition 4.7.** *For X a persistence module and $p \in (1,\infty)$,*

$$\int_\Omega \operatorname{rk} X_{s\leq t}\, g_p(s,t)\mathrm{d}s\mathrm{d}t = \|X\|_p^p.$$

Proof. We first consider the case where X is a single bar $K(a,b)$. Suppose that a and b are finite (i.e., not $-\infty$ nor $+\infty$, see 2.1). Then, for all $p \in (1,\infty)$, we compute

$$\int_\Omega \operatorname{rk} K(a,b)_{s\leq t}\, g_p(s,t)\mathrm{d}s\mathrm{d}t = \int_a^b \int_s^b p(p-1)(t-s)^{p-2}\mathrm{d}s\mathrm{d}t$$

$$
\begin{aligned}
&= \int_a^b p(b-s)^{p-1}\mathrm{d}s \\
&= (b-a)^p \\
&= \|K(a,b)\|_p^p.
\end{aligned}
$$

If a or b is not finite, then we still have the equality, as $\infty = \infty$.

The rank function $\operatorname{rk} X$ is the sum of the the rank functions of the bars in the bar decomposition of X. In order to generalize the previous result, it suffices to check that

$$\int_\Omega \sum_{i=1}^{\infty} \operatorname{rk} K(a_i,b_i)_{s\le t} g_p(s,t)\mathrm{d}s\mathrm{d}t = \sum_{i=1}^{\infty} \int_\Omega \operatorname{rk} K(a_i,b_i)_{s\le t} g_p(s,t)\mathrm{d}s\mathrm{d}t,$$

where the sums are over non-ephemeral bars, as ephemeral bars do not contribute to the integral.

◀

▶ **Theorem 4.8.** *Let $f\colon X \hookrightarrow Y$ be a monomorphism in* Pfd *and f_φ the bar-to-bar monomorphism induced by the canonical matching between X and Y (this assumes that we fix barcodes). Then, for all $p \in [1,\infty]$,*

$$\|\operatorname{coker} f\|_p \ge \|\operatorname{coker} \Delta f\|_p \ge \|\operatorname{coker} f_\varphi\|_p.$$

Proof. For $p \in (1,\infty)$, we combine Propositions 4.1 and 4.7, along with the fact that the function $h \mapsto \int_\Omega hg_p$ is increasing.

For $p = 1$ and $p = \infty$, we conclude by continuity of p-norms with respect to p. ◀

4.2 Recovering isometry result for Wasserstein distances

To have a definition of the algebraic Wasserstein distance in Pfd we should show that $\|\cdot\|_p$ (defined using the equality in Proposition 4.7) is a noise system on Pfd. We know it is true in Tame.

To prove the 'easy' axiom of noise systems we can proceed as follows, showing a stronger statement.

▶ **Proposition 4.9.** *For every short exact sequence $0 \to X \to Y \to Z \to 0$ it holds that $\operatorname{rk}(Y) \ge \operatorname{rk}(X) + \operatorname{rk}(Z)$.*

Proof. Let $s \le t$ in $\mathbb{R}$ and consider the appropriate diagram with two exact rows as in the snake lemma. We have

$$\operatorname{rk} X_{s\le t} - \operatorname{rk} Y_{s\le t} + \operatorname{rk} X_{s\le t} = -\dim\ker X_{s\le t} + \dim\ker Y_{s\le t} - \dim\ker Z_{s\le t} \le 0,$$

where the equality follows from $\operatorname{rk} X_{s\le t} = \dim X_t - \dim\ker X_{s\le t}$ and similar identities for Y and Z, together with the identity $\dim X_t - \dim Y_t + \dim Z_t = 0$ by exactness of $0 \to X_t \to Y_t \to Z_t \to 0$; the inequality above follows from the exact sequence $0 \to \ker X_{s\le t} \to \ker Y_{s\le t} \xrightarrow{g} \ker Z_{s\le t} \to \operatorname{coker} g \to 0$ (easy part of the snake lemma) again by Euler characteristic. ◀

By Proposition 4.7 and monotonicity of the integral, we have the following consequence of Proposition 4.9.

▶ **Corollary 4.10** ([14], Remark 7.32). *If $p \in [1, \infty)$ and $0 \to X \to Y \to Z \to 0$ is a short exact sequence, then*

$$\left(\|X\|_p^p + \|Z\|_p^p\right)^{\frac{1}{p}} \leq \|Y\|_p.$$

The 'easy' axiom of noise systems $\|X\|_p \leq \|Y\|_p$ and $\|Z\|_p \leq \|Y\|_p$ follows immediately. The case $p = \infty$ follows by taking the limit.

We have to prove the 'difficult' axiom of noise systems.

▶ **Proposition 4.11.** *For every short exact sequence $0 \to X \to Y \to Z \to 0$ it holds that $\|Y\|_p \leq \|X\|_p + \|Z\|_p$.*

Proof. We follow the proof strategy of [1, Lemma 4.15]. ◀

We provide an alternate (for one direction at least) proof to [14, Theorems 7.27-7.28].

▶ **Theorem 4.12.** *For any X and Y in Pfd, we have $d_p^p(X, Y) = W_p(\mathsf{Dgm}(X), \mathsf{Dgm}(Y))$.*

Proof. $\boxed{\leq}$ Let $\varphi\colon \mathsf{Dgm}(X) \to \mathsf{Dgm}(Y)$ be a matching. We want to construct a span with a smaller or equal cost. Following [14, Theorem 7.28], define

$$Z \coloneqq \bigoplus_{((a,b),(c,d)) \in \varphi} K(\max(a, c), \max(b, d))$$

and construct the natural span $X \leftarrow Z \to Y$. We can check that the cost of this span is exactly the cost of φ.

$\boxed{\geq}$ Let $X \xleftarrow{f} Z \xrightarrow{g} Y$ be a span. We want to construct a matching with a smaller or equal cost. First, factor the span as $X \hookleftarrow \operatorname{im} f \twoheadleftarrow Z \twoheadrightarrow \operatorname{im} g \hookrightarrow Y$ and replace each morphism with the morphism induced by canonical matchings. By Theorem 4.8 and its dual result, the cost of the new span $X \xleftarrow{f_b} Z \xrightarrow{g_b} Y$ is smaller than the cost of the original span.

The (strictly bar-to-bar) morphisms f_b and g_b induce matchings $\varphi_X\colon \mathsf{Dgm}(X) \to \mathsf{Dgm}(Z)$ and $\varphi_Y\colon \mathsf{Dgm}(Y) \to \mathsf{Dgm}(Z)$, respectively. Define

$$Z' \coloneqq \bigoplus_{\substack{(a,b) \in \mathsf{Dgm}(X) \\ (c,d) \in \mathsf{Dgm}(Y) \\ \varphi_X(a,b) = \varphi_Y(c,d)}} K(\max(a, c), \max(b, d)).$$

Observe that f_b and g_b factor through Z'. This defines a new span (of strictly bar-to-bar/algebraic matching morphisms) $X \leftarrow Z' \to Y$ with a smaller cost than the previous span: essentially, for each bar $\langle e, f\rangle$ in Z, we consider its matched bars $\langle a, b\rangle$ and $\langle c, d\rangle$ in X and Y, respectively, and we move the endpoints of $\langle e, f\rangle$ as far left as possible while keeping f_b and g_b well-defined. This makes the cokernels and kernels of these two morphisms smaller.

By the discussion for the $\boxed{\leq}$ case, the cost of this span is equal to the cost of the matching $\psi = \varphi_X \circ \varphi_Y^{-1}$. Thus $\psi\colon \mathsf{Dgm}(X) \to \mathsf{Dgm}(Y)$ is a matching whose cost is smaller than the original span. This concludes the proof. ◀

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